Safer3D: Multi-View 3D Object Detection for Safer AV Perception via Spatial Feature Refinement and Salient Query Propagation

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Abstract. In modern autonomous driving, the perception system is indispensable for accurately assessing the surrounding environment, needed for safe and accurate prediction and planning. In recent years, 3D object detection has been a reliable perception backbone for an autonomous vehicle (AV) system. However, considering the complex practical scenarios, accurate detection of small and occluded targets is a major challenge for autonomous driving with limited on-board computational resources. Existing state-of-the-art (SOTA) methods use a sparse query-based framework that leverages high-quality 2D object priors to generate 3D queries to improve recall for distant objects. We argue that this 2D prior-based supervision alone is insufficient to effectively detect small, occluded objects for safety-critical AV perception. In this work, we propose a safety-critical framework, Safer3D, that pivots on query-based multi-view 3D object detection with spatial feature upsampling and salient temporal object ranking to improve the detection of small and occluded objects. To effectively recognize distant small objects, we propose using super-resolution (SR) to zoom in on spatial features and sharpen object boundaries. We also propose a new size-adaptive query selection (SAQS) criterion to effectively select low-confidence small objects. Finally, to improve the recall on occulted and rare objects, we propose a query-based scene graph network for saliency-ranked query propagation (SRQP). Our proposed Safer3D demonstrated SOTA performance on Argoverse 2 and nuScenes benchmark datasets, over a range of up to 152 meters. Our proposed modules are dataset-agnostic and can be coupled with any query-based detection network to improve performance.

Keywords: Autonomous Vehicle Perception · 3D Multi-view Detection · Small Object Detection

1 Introduction

Long-range 3D object detection is a cornerstone of perception in autonomous vehicles (AVs): it defines the planning and control decisions and ultimately sets the limits of operational safety margins [10]. Yet, it remains one of the major challenges toward AV safety due to data sparsity and limited pixel information at a distance.



Distribution	(< 50m)	(≥ 50m)	(< 100m)	(≥ 100m)
Num. of obj (%)	0.462	0.537	0.792	0.207

Fig. 1: Challenging scenarios and GT distribution under different distance thresholds on Argoverse 2 validation set.

As shown in Figure 1, the "long-range" scenario is part of everyday driving, approximately 54% of objects lie beyond 50m and 20% beyond 100m. For vehicles moving at 130 km/h (80 mph), the system must detect obstructions at least 150 meters ahead to provide enough time for decision-making and safe braking [46]. The perception range must scale with speed to maintain a safe Time to Collision, typically aiming for a reaction time of 2.6 seconds or more. Thus, while $> 100\,\mathrm{m}$ is often labeled long-range in computer vision, on roadways it affords only $1.8s$ for perception and decision-making. According to [37], at 60 mph, the typical stopping distance for a vehicle is 73 meters, and it grows significantly under harsh road conditions, such as having a wet surface [13, 19].

Another noteworthy challenge for safe AV trajectory planning is detecting occluded objects at both near and far ranges, especially in city traffic scenarios. Detecting and avoiding collisions with Vulnerable Road Users (VRUs) and on-road objects, such as pedestrians, cyclists, traffic cones, traffic barrels, and road workers, is one of the numerous challenges of autonomous driving. Most of the time, the sidewalks are covered with billboards, electric or traffic poles, trees, and so on (See Figure 1), obscuring VRUs and other critical objects related to the safety measurement, making AV perception even more challenging [4, 43].

To tackle the far-object detection challenge, several recent works propose using a camera-based multi-view perception system [14, 36, 39] due to its low deployment cost, widespread applicability, and ability to leverage comprehensive context information from the ego scene. Sparse query-based surround-view methods are computationally efficient and effective for sparse scenes. It utilizes fixed global queries to represent 3D objects [5, 36]. The fixed global queries lack generalization for dynamic scenarios and often fail to detect long-range objects. To address this issue, Jiang et al [14] introduce adaptive queries from a 2D region proposal head and accompany them with the fixed global queries. However, none of the works ensures the quality of the feature from which the query will be chosen. Also, the query selection in these works treats the entire scene uniformly; hence, small objects in the far range are often excluded from the query set due to low confidence. Another challenge in safe AV trajectory planning is accurately detecting occulted objects in a complex, incomplete scene. The current work uses temporal object propagation to retain objects from earlier scenes [18, 33]. Most existing temporal propagation strategies prioritize propagating high-confidence object queries, rather than explicitly preserving and updating far-range, isolated, or partially occluded objects. As a result, object queries that require greater attention during temporal propagation are often dropped early, undermining the

intended role of query propagation. This underscores the importance of far-range detection for AV safety margins.

To address the issues outlined above, we first enhance the quality of spatial information in low-receptive field (LRF) features by applying super-resolution (SR) from high-receptive field (HRF) features. At higher strides, the objects in the $> 100m$ range become tiny (Figure 1) and the object boundary blends into the background scene. Both conditions cause the 2D detector to miss far objects, ultimately degrading far-object recall. With the help of SR feature refinement, we generate a higher-quality 2D prior than other SOTA approaches, thereby improving localization accuracy and recall. Next, existing work [14,35] performs adaptive query selection from a 2D prior solely based on the objectness score; however, we propose incorporating object size into the selection process. With this, we will be able to retain objects that are slightly less confident; however, they are good candidates for small objects. Finally, to improve AV perception performance for VRUs and safety-critical objects, we proposed a saliency-ranked query propagation (SRQP) module. The goal of this module is to treat each object query as a node and create a graph attention network. The graph attention network increases the saliency score of occluded or rare objects in the scene by leveraging geospatial attention, query confidence, and contextual information. The saliency information is later used to rank the most salient object in the current scene. Each of these modules helps the Safer3D framework to focus on small and occulted objects and improve the safety of the AV perception system. Our proposed method achieves remarkable performance at long range on the Argoverse2 dataset, outperforming existing Camera/LiDAR-based SOTA methods by a significant margin. We have also evaluated our Safer3D model on the challenging nuScenes dataset to validate its SOTA performance and generalization capabilities. Our contribution can be summarized as follows:

1. We introduce a novel feature enhancement module to generate a high-quality 2D prior for 3D object detection and to improve recall for distant objects.
2. The effectiveness of query-based detection frameworks largely depends on selecting effective object queries. Motivated by this, we introduce Size-Adaptive Query Selection (SAQS), a size-aware criterion that explicitly incorporates object scale into the query selection process.
3. For improved performance of safety-critical VRUs and occluded objects, we propose a graph attention network (SRQP) that calculates the object saliency score based on the graph's semantic and geometric context.
4. Our Safer3D framework sets a new SOTA on Argoverse2 and nuScenes benchmark datasets, with particularly strong gains in recall.

2 Related Works

2.1 Multi-view 3D object detection

A common challenge is reconciling diverse viewpoints into a unified 3D representation while handling occlusion and depth ambiguity. Early studies emphasize ef-

ficient multi-view feature transformation to a shared 3D space or a unified coordinate frame, enabling 3D reasoning from camera observations [12,15,22,26,35,36]. Existing multi-view methods can be broadly categorized into BEV-based and sparse query-based paradigms. BEV-based approaches project multi-camera features into a bird's-eye-view representation and then apply a 3D detection head in BEV space, leveraging estimated depth to construct BEV features and predict object [11,12,21,22]. Some works further incorporate spatiotemporal interaction within BEV by using grid-shaped BEV queries that attend to spatial and temporal cues [22]. Others improve robustness by introducing camera-aware depth estimation modules to obtain more reliable BEV features [21]. In the multi-view temporal setting, sequential modeling and Long-term temporal propagation have also been investigated via temporal self-attention, adjacent-frame BEV fusion to alleviate occlusion and stabilize detection across dynamic scenes [11, 20, 31].

In contrast, sparse query-based methods avoid explicitly forming dense BEV maps and instead learn a compact set of global object queries from representative features. This line of work is often computationally lightweight and naturally supports object-centric reasoning at inference time [26, 35]. However, temporal modeling becomes less straightforward without an intermediate dense representation, and some approaches introduce global cross-attention or sparse attention mechanisms, potentially increasing computational overhead [24, 27, 28]. Despite this, sparse queries are well-suited for representing moving objects, since query states can be propagated and updated over time [24].

2.2 Query Propagation

Query propagation has emerged as a lightweight way to inject temporal continuity into transformer-based perception by treating queries as persistent object states rather than re-computing dense temporal features. Since DETR [5] has been published, many downstream tasks have used query as object to model temporal interaction between objects [9, 29, 42, 44, 45]. To reduce the computational cost on non-key frames, QueryProp [9] performs query interaction. A similar idea is used by 3D-MAN [40], effectively combining different perspective frames from the same scene to extend a single-frame LiDAR detector into a multi-frame LiDAR detector. Built upon the sparse query design in the PETR [26,27] series, StreamPETR [33] develops an object-centric temporal mechanism that propagates long-term historical information online through object queries frame by frame. LWDN [16] incorporates a brain-inspired memory mechanism to propagate and update the query feature from video data. BEVDet4D [11] extends the BEVDet paradigm from the spatial-only 3D detector to the spatial-temporal 4D detector by fusing BEV features with the previous frame. Query propagation has also been very popular in object-tracking tasks. MOTR [42] and TransTrack [32] propose tracking the query to model object association across the temporal dimension. To further boost performance, [30, 42, 44] introduce a 3D track query to model spatial and appearance coherent tracks for each object that appears in multiple cameras and multiple frames. MeMOT [3] builds a memory bank to model long-term object dependencies. On the other hand, CoopTrack [47]

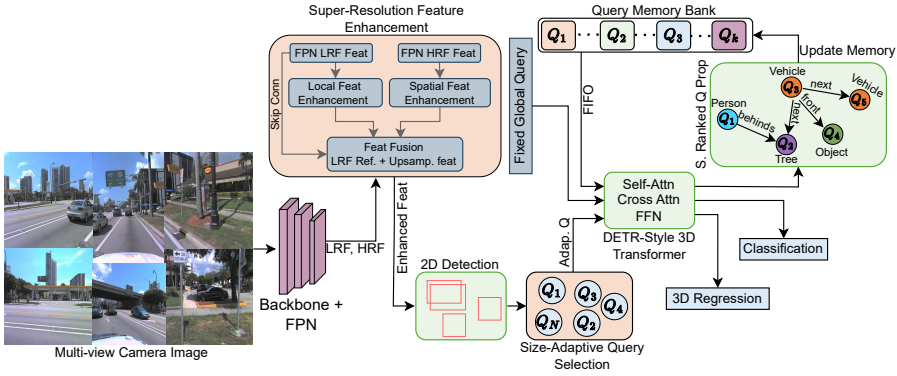


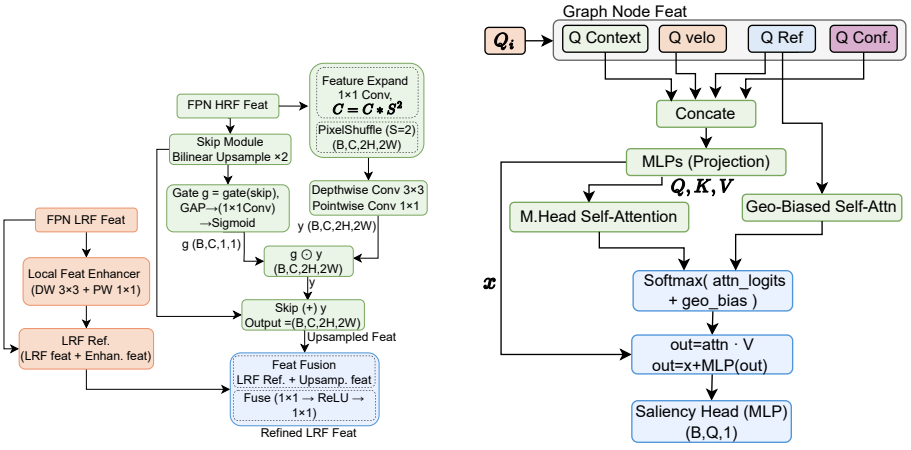
Fig. 2: The Overall Architecture of the proposed Safer3D model. The multi-scale features from FPN are refined to enhance the spatial semantics. The 2D proposals from the 2D detector are converted into 3D proposals with depth estimation. The SAQS module selects candidate queries from 3D proposals. The final set of queries is then curated using a scene-graph-based saliency ranking for effective query propagation.

transmits sparse instance-level features that significantly enhance perception capabilities while maintaining low transmission costs.

3 Methodology

The overall architecture of the Safer3D framework is illustrated in Figure 2. Safer3D accepts multi-view camera inputs from a particular scene. The input for a scene S_i can be denoted as $S_i = \{I_1, I_2 \dots I_N\}$, where I_i denotes a perticular image and N denotes the number of cameras. The images from the scene are fed into the feature extractor, followed by an FPN that generates multi-scale image features, $F_i = F^1, F^2, \dots, F^l$, where l is the number of feature scales. The multi-scale features are then fed into a super-resolution-based feature enhancement module to improve feature quality for small and distant object detection.

The 2D detection head uses enhanced features and generates 2D proposals for each feature scale. The detection head also employs a depth estimator (e.g., DepthNet) to estimate depth for each 2D proposal. The 2D proposals are then filtered by the size-adaptive query selection module and projected into 3D space to generate 3D object queries. These adaptive 3D queries are finally concatenated with fixed global queries and temporally propagated queries from the memory bank and fed into the DETR-style [35] transformer decoder for context-aware feature generation for prediction and regression tasks. Finally, we introduce a novel scene graph-based network where each object query is treated as a node in the graph, and we calculate the saliency score for each query based on the spatial relation and scene context. The top-k saliency-ranked adaptive queries are propagated to the memory bank to improve the performance for partially visible objects and incomplete scenes in future frames.



(a) Super-Resolution based Feature Enhancement (SRFE)

(b) Saliency-Ranked Query Propagation (SRQP)

Fig. 3: (a) SRFE restores fine spatial cues by enhancing HRF features and refining LRF features. (b) SRQP ranks queries via geometry-biased self-attention to propagate salient (often occluded) instances.

3.1 Super-Resolution based Feature Enhancement

To improve performance at sparse ranges, recent works [1, 14, 33, 39] use 2D priors to generate 3D proposals. Motivated by this observation, we also leverage a 2D prior to generate 3D proposals for improved small-object detection. Although the existing 2D prior-based 3D detection module achieved superior performance compared with LiDAR-based methods, these works overlook the pixel-distribution statistics of sparse range objects. The sparse range object can span as few as ≈ 10 pixels [14,39]. At higher striding, the semantic details of small objects are washed out, and the object boundary blends with the scene background. In both scenarios, 2D priors are not very helpful for generating reliable 3D queries. To resolve this issue, we propose enhancing the low-receptive-field (LRF) feature by combining the high-receptive-field (HRF) feature (as illustrated in Figure 3a) through feature expansion and gated feature fusion.

Starting from the high receptive-field feature $\mathbf{F}^{\mathrm{HRF}} \in \mathbb{R}^{B\times C\times H\times W}$ from FPN, we first construct a stable upsampled baseline ($\mathbf{S} \in \mathbb{R}^{B\times C\times sH\times sW}$) using bilinear interpolation with scale (s=2), which preserves global semantics and provides a reliable reference at higher spatial resolution.

In parallel, we learn an SR residual that explicitly learns to restore spatial details lost at coarse stride. We first apply a 1×1 conv. to expand HRF channels into s^2 sub-pixel groups (Figure 4),

$$\mathbf{Y}_0 = \phi^{\exp}_{1\times 1}(\mathbf{F}^{\mathrm{HRF}}) \in \mathbb{R}^{B\times(s^2C)\times H\times W}, \tag{1}$$

and then use PixelShuffle to rearrange these groups into a higher-resolution map denoted as $\mathbf{Y_1} \in \mathbb{R}^{B\times C\times sH\times sW}$. Unlike bilinear interpolation, PixelShuffle does

not smooth features; instead, it converts the learned s^2 channel groups at each location into an $s \times s$ spatial pattern (e.g., four sub-pixel positions when s=2). This enables a far-range object that would otherwise collapse to a single HRF cell to produce a structured local response at higher resolution, preserving boundary and localization cues that are critical for generating reliable 2D priors.

To suppress upsampling artifacts and sharpen local structures with minimal overhead, we refine the sub-pixel output using a lightweight depthwise-separable block, $\mathbf{Y} = \phi^{\mathrm{pw}}_{1\times1}\big(\phi^{\mathrm{dw}}_{3\times3}(\mathbf{Y}_1)\big)$. The depthwise 3×3 convolution enforces local spatial consistency at negligible cost, which is particularly effective for removing sub-pixel rearrangement noise introduced by PixelShuffle and point-wise 1×1 mixes recovers discriminative features for recognition. Since aggressive SR residuals can also introduce noisy patterns in sparse/low-texture regions, we further compute a channel-wise gate from the upsampled baseline (S) to control how much SR detail should be injected:

$$\mathbf{g} = \sigma\big(\phi^{g}_{1\times1}(\mathrm{GAP}(\mathbf{S}))\big) \in \mathbb{R}^{B\times C\times1\times1}, \qquad \widetilde{\mathbf{F}}^{\mathrm{HRF}} = \mathbf{S} + \mathbf{g} \odot \mathbf{Y}. \tag{2}$$

This gated aggregation retains the robust semantics of $\mathbf{S}$ while selectively adding discriminative details from $\mathbf{Y}$. Finally, we fuse the locally enhanced low receptive feature ($\mathbf{F}^{\mathrm{LRF}}$) with the SR-enhanced HRF feature to obtain the refined representation:

$$\widetilde{\mathbf{F}}^{\mathrm{LRF}} = \mathbf{F}^{\mathrm{LRF}} + \phi^{\mathrm{loc}}\big(\mathbf{F}^{\mathrm{LRF}}\big), \qquad \mathbf{F}^{\mathrm{ref}} = \phi^{\mathrm{fuse}}\Big(\widetilde{\mathbf{F}}^{\mathrm{LRF}} + \widetilde{\mathbf{F}}^{\mathrm{HRF}}\Big), \tag{3}$$

where ϕ^{fuse} is implemented as a simple convolutional operation followed by ReLU activation. This fusion injects HRF semantics at a higher effective spatial resolution while preserving the fine localization cues of LRF features, thereby improving the reliability of priors/queries for small and distant objects.

3.2 Size-adaptive Query Selection

The enhanced features from FPN are fed into the YOLOX-style 2D detection head. The 2D detection head provides an objectness score and a depth estimate for each location in the dense BEV feature. Previous work [14, 39] selects 2D priors for query generation based on the objectness scores of 2D proposals from the YOLOX head. However, it is evident from Figure 1 that the distant objects appear small in size and may receive a low objectness score in comparison to a much closer, clearly visible object. Hence, relying solely on objectness confidence would effectively reject small objects and reduce overall recall. To address this challenge, we propose computing a smallness prior from each 2D proposal generated by the 2D YOLOX head. The complete set of operations for size-adaptive query generation is provided below:

Given the decoded 2D boxes $\mathbf{b}_i = (x_i^1, y_i^1, x_i^2, y_i^2)$, we first compute the proposal area (A_i) with width and height as $(x_i^2 - x_i^1)$ and $(y_i^2 - y_i^1)$, respectively, and apply clamping to avoid negative values that can arise from imperfect regression. To make the prior comparable across different input resolutions, we prepare a normalized proposal area $\widetilde{A}_i$ by the image area A_{img} (with a small ϵ for numerical stability).

This normalization ensures that the same physical object occupying a similar fraction of the image receives a similar prior, independent of absolute pixel dimensions. We then map the normalized area to a *smallness prior* using an exponential kernel,

$$\pi_i^{\text{small}} = \exp\left(-\lambda \widetilde{A}_i\right), \qquad (4)$$

where λ controls how strongly large boxes are down-weighted. This function is strictly monotonic and smoothly bounded, preserving small proposals while preventing very large near-range boxes from dominating the ranking. Finally, we combine this prior with the base objectness-score from the YOLOX head (denoted as ρ_i^{obj}) to obtain the query selection score, $Qscore_i = \rho_i^{obj} \cdot \pi_i^{\text{small}}$, where ρ_i^{obj} enforces standard confidence filtering (removing clear negatives) and π_i^{small} performs size-adaptive re-ranking among the remaining candidates. In this way, small distant objects that would otherwise be suppressed by objectness-only selection are retained for query generation.

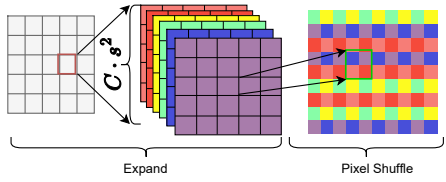


Fig. 4: Sub-pixel feature Upsampling. A lightweight channel-expansion layer prepares the HRF feature for PixelShuffle, which rearranges channel groups into a higher-resolution feature map, enabling learned spatial detail recovery.

3.3 Saliency-ranked Query Propagation

After selecting adaptive queries by thresholding $Qscore_i$, the proposed 3D adaptive queries are concatenated with initialized, fixed global queries and fed into subsequent transformer layers in the decoder. Inspired by [14], we adopt a decoder architecture in which perspective-aware aggregation enables efficient feature interactions across different scales and views. Features from multi-view, multi-scale image planes are aggregated into 3D queries according to their learned importance, and the resulting contextualized queries are passed to the classification and regression heads for final prediction.

As discussed earlier, a major challenge in urban driving scenes is the partial visibility of VRUs and construction-related objects on the road, both of which are critical to safety. Prior works improve occluded-object detection by propagating top-K queries from the previous frame to the next, typically ranked by classification confidence. However, a high classification score neither indicates partial visibility nor guarantees that propagated queries are non-overlapping or represent a rare object. To address this limitation, we propose *saliency-ranked query propagation* (SRQP) (Figure 3b), which learns query importance by constructing a fully-connected scene graph over queries and estimating a per-query saliency score via geometry-aware interactions. Each query is treated as a node, with attributes including the query context feature, 3D reference, velocity, and confidence score.

Formally, let $\mathbf{M} \in \mathbb{R}^{B\times Q\times C}$ denote query context features, $\mathbf{P} \in \mathbb{R}^{B\times Q\times 3}$ the 3D reference points, $\mathbf{V} \in \mathbb{R}^{B\times Q\times 2}$ the planar velocities, and $\mathbf{p}^{\text{det}} \in (0,1)^{B\times Q\times 1}$

the query confidence. We first fuse these heterogeneous cues into a compact node ($\mathbf{U} = \left[\mathbf{M}; \mathbf{P}; \mathbf{V}; \mathbf{p}^{\text{det}}\right]$), and turn it into a unified feature ($\mathbf{X} = f_{\text{in}}(\mathbf{U}) \in \mathbb{R}^{B\times Q\times d}$) compatible for attention operations.

Initially, we create a fully connected graph by applying multi-head self-attention to the set of queries to model pairwise interactions and estimate query relevance. The attention logits can be denoted as $\mathbf{L} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_h}}$.

Next, to model occlusion scenarios, we use the edge geometry mechanism. We compute the 3D offsets between each pair to determine the relative positions of the queries, and compute the BEV distance between them. With this, we determine whether an object is in front of another and whether they are very close or overlap. We also get benefited by the confidence score injected as a node attribute. A confidence score helps draw attention to a low-confidence object that is obscured by a high-confidence object. The context information at each node helps identify redundant object queries based on feature similarity, thereby rescuing rare/isolated objects by increasing their saliency. The relative-geometry bias for each pair (i, j) is computed as follows:

$$\Delta_{ij} = \left[\, \Delta x_{ij}, \Delta y_{ij}, \Delta z_{ij}, \Delta r^2_{ij} \,\right], \quad \Delta r^2_{ij} = \Delta x^2_{ij} + \Delta y^2_{ij}, \quad B = f_{\text{geo}}(\Delta_{ij}), \quad (5)$$

The resulting attention weights are computed as $\mathbf{A} = \text{softmax}(\mathbf{L} + \mathbf{B}) \in \mathbb{R}^{B\times H\times Q\times Q}$. The updated node representation is obtained by attention aggregation followed by an output projection and a residual connection:

$$\mathbf{O} = \mathbf{A}\mathbf{V}_a \in \mathbb{R}^{B\times H\times Q\times d_h}, \qquad \hat{\mathbf{X}} = f_{\text{out}}(\mathbf{O}) + \mathbf{X} \in \mathbb{R}^{B\times Q\times d}, \quad (6)$$

Finally, we predict a per-query saliency logit $\mathbf{s}^{\text{sor}} = \boldsymbol{\ell}^{\text{det}} + f_{\text{sal}}(\hat{\mathbf{X}})$ using a compact MLP head. Here, $\boldsymbol{\ell}^{\text{det}}$ denotes the detection logit used for objectness guidance. Next, a sigmoid function is applied to rank and propagate the top-K salient queries in the memory bank. In contrast to confidence-only propagation, SRQP favors spatially isolated or partially observed instances (e.g., occluded VRUs) and down-weights redundant candidates through geometry-aware interactions, thereby improving temporal continuity and recall under occlusion.

Loss Calculation. To optimize the saliency ranking module, we create target labels for each query/node in the scene graph. We perform one-to-one Hungarian matching between predicted queries $(\mathbf{b}, \mathbf{c})$ and ground-truth $(\mathbf{g}, \mathbf{y})$ using the matching cost, and $\varnothing$ denotes an unmatched (background) query.

$$\pi^* = HungAssign(b, c, g, y), \qquad y_i^{\text{sor}} = \begin{cases} 1, & \pi^*(i) \neq \varnothing, \\ 0, & \text{otherwise}, \end{cases} \quad i = 1, \ldots, Q. \quad (7)$$

After getting target labels y_i^{sor}, we use a BCE loss function (eq. 8) to calculate the loss between node saliency and node label. This loss is combined with the classification and regression losses and is jointly optimized at each iteration.

$$\mathcal{L}_{\text{sor}} = \frac{1}{BQ} \sum_{b=1}^{B} \sum_{i=1}^{Q} \text{BCEWithLogits}\left(s_{b,i}^{\text{sor}},\, y_{b,i}^{\text{sor}}\right), \quad (8)$$

4 Experiments

We evaluate our Safer3D model on two popular benchmark datasets for 3D object detection. The first benchmark dataset is Argoverse2 [38], which is widely used for perception and prediction tasks. The Argoverse2 dataset comprises 1000 driving scenes, each 15 seconds long. Among them, 700 were kept for training, 150 for validation, and 150 for testing. The 7-ring cameras mounted on the vehicle provide 360° horizontal FOV. This dataset comprises 26 object classes, with objects ranging up to 152m.

Another popular 3D object detection dataset for autonomous scenes is nuScenes [2]. This dataset provides 1000 driving scenes, each lasting 20 seconds. The annotations contain up to 1.4M 3D bounding boxes with long-range object annotation up to 80m. This dataset covers 10 object classes. Each scan consists of 6 surround-view images, providing a total 360° horizontal FOV.

To evaluate the performance, we compare our Safer3D model with several state-of-the-art methods. We follow standard performance metrics such as mAP and Recall, with three error metrics: ATE, ASE, and AOE. To better compare with other methods, we also include the official Argoverse2 metrics, the "Composite Detection Score (CDS), and the nuScenes Detection Score (NDS). As one of our primary goals is to improve VRU safety, we present a class-wise Average Precision (AP) performance evaluation for VRU classes. The class-wise performance will demonstrate Safer3D's improvement on VRU classes and highlight the need for greater attention to these classes in a safer AV perception system.

4.1 Implementation Details

The Safer3d implementation relies on StreamPETR [33] and Far3D [14] code base. Following prior work, we use the VoVNet-99 [17] model pre-trained with FCOS3D [34] on the nuScenes dataset. We also explored COCO [23] prerained ViT-Large [7] as a backbone for improved performance. The FPN provides multi-scale features with strides of 1/8, 1/16, 1/32, and 1/64. For model optimization, we use AdamW with a weight decay of 0.01. The batch size and learning rate is set 4 and 2e-4, respectively. The number of epochs is set according to the training settings used in the baseline. All images in the Argoverse2 dataset were first resized to a consistent resolution and then underwent the same image augmentation for each camera view. The other settings remain consistent with the Far3D configuration.

4.2 Main Results

Argoverse 2 SOTA Performance. In Table 1 we present comprehensive performance analysis on Argoerse2 val set comparing several latest SOTA methods. Early BEV methods, such as BEVStereo and SOLOFusion, struggle to achieve satisfactory performance with camera images. The mAP for these methods was approx. 15%, showing infeasibility for a safer AV perception system. The

Table 1: Comparisons on the Argoverse 2 val set. We evaluate 26 object categories with a range of 152 meters. Safer3D outperforms previous SOTA methods with a large margin on mAP and CDS performance metrics.

Methods	Backbone	Modality	mAP↑	CDS↑	mATE↓	mASE↓	mAOE↓
BEVStereo [20]	VoV-99	Camera	0.146	0.104	0.847	0.397	0.901
SOLOFusion [31]	VoV-99	Camera	0.149	0.106	0.934	0.425	0.779
PETR [26]	VoV-99	Camera	0.176	0.122	0.911	0.339	0.819
Sparse4Dv2 [25]	VoV-99	Camera	0.189	0.134	0.832	0.343	0.723
StreamPETR [33]	VoV-99	Camera	0.203	0.146	0.843	0.321	0.650
Far3D [14]	VoV-99	Camera	0.244	0.181	0.796	0.304	0.538
Safer3D (Ours)	VoV-99	Camera	0.267	0.215	0.728	0.277	0.480
CenterPoint [41]	-	Lidar	0.274	0.210	0.548	0.362	0.781
FSD [8]	-	Lidar	0.291	0.233	0.468	0.299	0.740
VoxelNeXt [6]	-	Lidar	0.307	0.225	0.431	0.291	1.157
Far3D	ViT-L	Camera	0.316	0.239	0.732	0.303	0.459
Safer3D (Ours)	ViT-L	Camera	0.327	0.243	0.683	0.282	0.568

performance significantly improved when multi-view sparse query-based methods, such as StreamPETR and Far3D, were introduced. The increase in mAP was about $\approx 10\%$, compared to the BEV methods, drawing attention to the use of a multi-view method with temporal propagation. Our proposed Safer3D framework with the VoV-99 backbone outperforms the previous SOTA methods by maintaining a minimum margin of $+\Delta 2\%$ in mAP and $+\Delta 3\%$ in CDS. We also reduced positional error for small and distant objects, minimizing mATE and mASE by 7% and 3%, respectively.

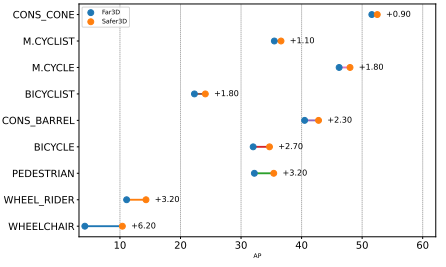


Fig. 5: Classwise AP comparison between the baseline (Far3D) and proposed Safer3D model for AV2 VRU and safety critical classes.

Additionally, we compared Safer3D against competitive LiDAR baselines, including CenterPoint and VoxelNeXt. VoxelNeXt achieves 30.7% mAP and 23.3% CDS, but our camera-only model with a ViT-L backbone performs better overall. Safer3D achieves 32.7% mAP and 24.3% CDS, beating the best LiDAR result and Far3D by +2% and +1.3% of mAP, respectively. In terms of the error metrics, the LiDAR models exhibit smaller localization errors due to the sensor's direct depth information. However, Safer3D does especially well on orientation, mostly because multi-view images provide richer scene context that helps reduce yaw ambiguity even without explicit LiDAR depth.

Table 2: Safer3D performance comparison with SOTA methods on the nuScenes `val` splits. The Safer3D performance on the nuScenes dataset shows model generalizability across different datasets.

Methods	Backbone	Modality	mAP↑	NDS↑	mATE↓	mASE↓	mAOE↓
PETR	ResNet101	Camera	0.366	0.441	0.717	0.261	0.412
SOLOFusion	ResNet101	Camera	0.483	0.582	0.503	0.264	0.381
StreamPETR	ResNet101	Camera	0.504	0.592	0.569	0.262	0.315
Sparse4Dv2	ResNet101	Camera	0.505	0.594	0.548	0.268	0.348
Far3D	ResNet101	Camera	0.510	0.594	0.551	0.258	0.372
Safer3D (Ours)	ResNet101	Camera	0.522	0.605	0.540	0.255	0.351
Safer3D (Ours)	ViT-L	Camera	0.528	0.611	0.543	0.249	0.363

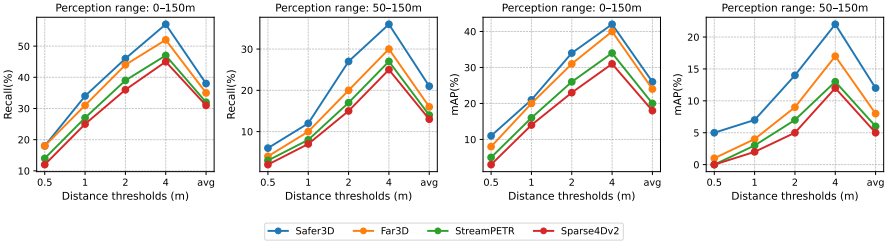


Fig. 6: Recall and AP of competitive methods with different distance thresholds. Metrics across different ranges show that our approach consistently achieves better results.

One of our main focuses is improving roadway safety for vulnerable road users (VRUs); hence, we report class-wise gains in safety-critical categories. Figure 5 summarizes the Average Precision (AP) of Far3D and Safer3D on several challenging VRU and traffic-safety classes. Safer3D (with ViT-L) consistently outperforms the baseline across all critical classes. Notably, we observe gains of +3.20% AP for Pedestrian, +2.70% for Bicycle, and +2.30% for Construction Barrel. We also observe clear improvements for Motorcycle (+1.8%) and Construction Cone (+0.90%), indicating improved detection of small, safety-relevant objects in cluttered scenes. Performance on Wheelchair and Wheeled Rider remains relatively low. These categories are both rare in the dataset and often appear under heavy occlusion, which limits the amount of clean supervision available during training and makes them difficult for perception.

nuScenes SOTA Performance. To prove model generalization, we provide cross-dataset performance on the nuScenes validation set. In Table 2, we present the Safer3D performance against camera-based SOTA methods. We validate Safer3D's performance using the ResNet101 and ViT-L backbones. It is evident that our Safer3D maintains superior performance, achieving an mAP of 52.8% and NDS of 61.1% with ViT-L. The mAP of Safer3D is 1.8% higher than that of Far3D, indicating that our saliency-ranked object temporal modeling outperforms Far3D both in accuracy and localization metrics.

Table 3: Ablation study on each component of the Safer3D model for Argoverse2 `val` set. We performed ablation analyses, both isolated and combined, using mAP and CDS metrics.

SRFE	SAQS	SRQP	mAP ↑	CDS ↑
			0.244	0.181
✓			0.249	0.188
	✓		0.251	0.187
✓	✓		0.257	0.201
✓	✓	✓	0.267	0.215

Table 4: Ablation for smallness prior hyperparameter (λ) on Argoverse2 dataset. This λ parameter controls the penalty to be imposed on the size of the object.

λ	mAP	Recall
0.5	0.251	0.365
1.5	0.262	0.380
1.7	0.267	0.384
2.0	0.255	0.369

4.3 Ablation Study

This section presents an ablation study to assess the effectiveness of the proposed modules on the Argoverse2 validation set. Starting from our baseline model, we isolate the improvements of the *Safer3D* modules and explain the rationale for each module by analyzing its performance and impact on the overall modeling process. For the ablation study, we will use the VoV-99 as the backbone to ensure a fair comparison with most camera-based models.

Table 3 begins with the performance of the Far3D baseline model. Next, we integrate a very lightweight feature enhancement module (SRFE) into the Far3D pipeline and achieve mAP and CDS gains of 0.5% and 0.7%, respectively. The learnable Param. incurred by SRFE is approx. $0.597M$. We also observe the isolated performance of the SAQS module ($\lambda = 1.7$), and achieve an improvement of $\approx 0.7\%$ on each metric. Next, we combine the SRFE and SAQS modules, achieving significant performance improvements of $+\Delta 1.3\%$ in mAP and $+\Delta 2\%$ in CDS. This aligns with the motivation for feature enhancement: if object-level semantic features are missed during feature extraction, the 2D detector will fail to generate high-quality object queries for small objects. Finally, to improve recall and mAP on occluded objects, we integrate the saliency-ranked query propagation module (SRQP), with a negligible number of learnable parameters ($\approx$ 0.125M). Figure 7 shows the qualitative performance of SRQP, where we plot the saliency score for a few candidate queries for propagation. It is evident that our scene graph learned to upweight saliency for occluded and isolated objects while preserving saliency for more visible objects.



Fig. 7: Safer3D saliency score for a few candidate queries learned from the Scene-graph module. The yellow circle denotes the query center.

We further investigate the impact of the smallness prior by varying the weighting factor λ. As reported in Table 4, the best performance is obtained with $\lambda = 1.7$, achieving an mAP of 0.267 and a recall of 0.562. When λ is reduced to

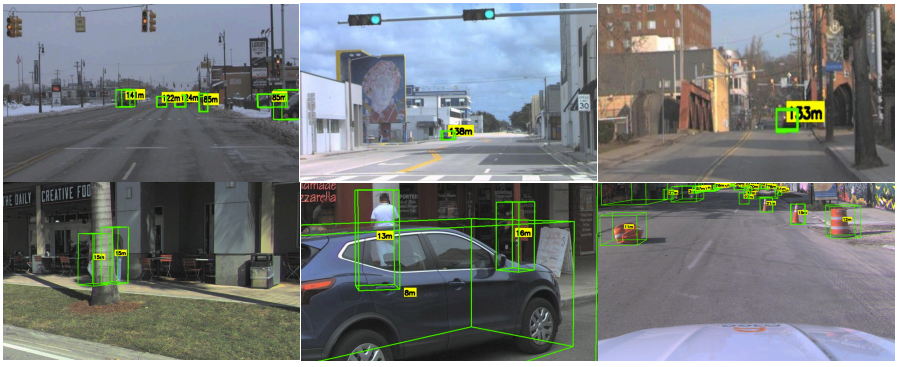


Fig. 8: Safer3D detection results on Argoverse2 dataset. The top row shows the detection for the far range, whereas the bottom row shows occluded and construction objects on the road. The object distance from the ego-vehicle is labeled in yellow.

0.5, the area prior becomes weak and yields only a marginal recall improvement over purely objectness-driven query selection. In contrast, setting $\lambda = 2.0$ overpenalizes large-area queries, which suppresses informative hypotheses and leads to a drop in overall mAP. Figures 6 provide quantitative evidence of Safer3D's behavior over the full perception (0–150 m) and the far (50–150 m) range. We see that Safer3D consistently outperforms prior methods and improves recall by approximately 4% over the Far3D baseline, with the largest gains observed in the 50–150m setting. These results indicate that SRFE and SAQS improve sensitivity to small foreground targets at long distances while maintaining robust detection in the overall range. Figure 8 further illustrates the detection performance of Safer3D in challenging situations. The top row shows the far-range detection capability, whereas the bottom row shows the detection performance for partially visible objects that need more attention from query propagation.

5 Conclusion

For a safe AV perception system, detecting small and occluded objects is critical. In this work, we argued that existing methods are not optimized for VRU classes and lack a sound rationale for adaptive query selection and query propagation. To address these issues, we proposed a super-resolution-based feature enhancement technique that is effective for small objects. Next, we propose a more appropriate criterion for adaptive query selection that incorporates smallness priors. Finally, we introduce a scene graph network that computes saliency scores using geometric attention among nodes. Experimental results on the Argoverse2 dataset demonstrate that Safer3D achieves consistent performance gains across both far and occulted scene scenarios. By detecting distant objects earlier, we extend the AV's planning horizon—giving the controller more time to make decisions and execute safer maneuvers. Future work will focus on safer AV perception under adverse weather conditions, where only camera methods may fail due to visibility limitations and camera sensor degradation.

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